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DELIVERABLE

D6.8 – Procedural Ocean Breakdown

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Executive Summary

This document offers a comprehensive overview of the UI and UX design of the Procedural Ocean platform, providing detailed explanations of all interface elements and their functionalities.

For visual reference please see the following two videos:

[Procedural Ocean \(youtube.com\)](#)

Video 1: Procedural Ocean Trailer

[Immersive Ocean Digital Twin Using Procedural Content Generation in Unreal Game Engine \(youtube.com\)](#)

Video 2: Webinar on using procedural content generation for Procedural Ocean

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Document History			
Version	Date	Contributor(s)	Description
1.0	09/07/2024	BUas	Finished PC & VR build of PO

Definitions, Acronyms and Abbreviations

Acronym/ Abbreviation	Title
BUas	Breda University of applied sciences
UI	User interface
PO	Procedural Ocean

1 UI/UX Breakdown

1.1 Main Menu

The main menu provides the following three options:

Scene

- This option presents a drop-down menu.
- Use the drop-down menu to select the desired JSON data file that you wish to load.
- The selected file determines the area or scene that the application will utilize.

Start

- Once the preferred JSON data file is selected from the Scene drop-down menu, click the Start button.
- This action initiates the loading process of the selected data file.

Quit

- Selecting this option will terminate the application.

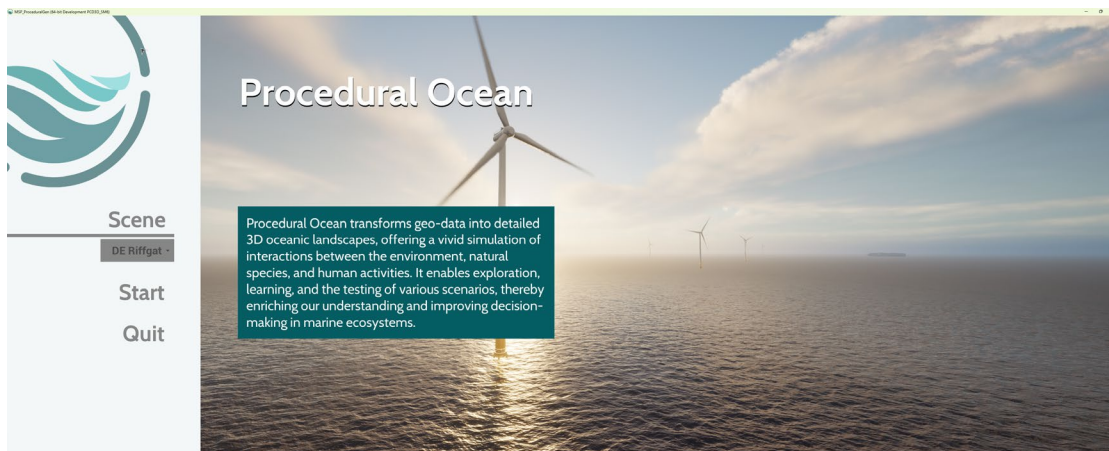


Figure 2: Main menu of the Procedural Ocean platform

1.2 Game Controls

The controls can be accessed through the settings menu at any time. Additionally, the controls will be displayed the first time the application is launched.

1.2.1 Controls Breakdown

- Auto Hide UI Bar: Allows you to hide the UI for screen captures.
- Close Game (without using the menu): Press **Alt + F4** to exit the application.

Movement Controls

- Move Forward, Left, Right, Backwards: Use the **W, A, S,** and **D** keys respectively.
- Move Straight Up: Press the **Spacebar**.
- Move Straight Down: Press the **Ctrl** key.
- Free Look Around: Hold down the right mouse button and move the mouse.

Interaction Controls

- Select an Object: Left-click on the object.
- Interact with an Object: Double-click the left mouse button or the middle mouse button, depending on the object.

Additional Functional Controls

- Speed Up/Slow Down: Hold down the right mouse button and scroll up to speed up or scroll down to slow down.
- Focus on an Object: Press the **F** key.

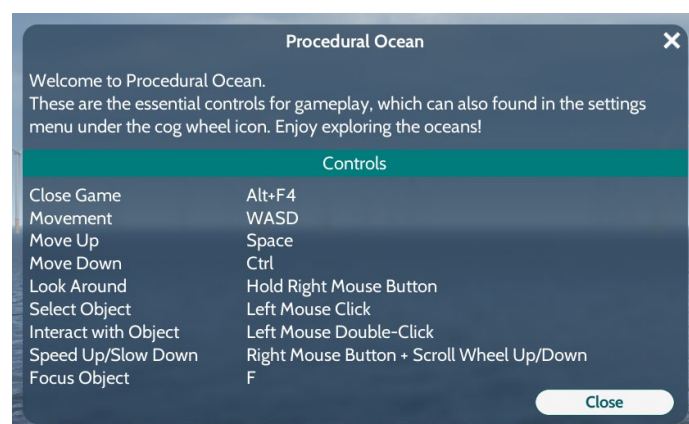


Figure 2: Tutorial window which will be prompted on the first time launching the Procedural Ocean platform.

1.2.2 Virtual Reality Mode

Connect your Meta Quest 3 headset to the PC. Then, launch the **pov.launcher.exe** application and select option '**4. Launch in VR mode**'. You will be prompted to select a JSON file; choose the desired file and press Enter.

Disclaimer: The weather controls + minimap are disabled in VR mode to save performance, ensuring a better player experience.

General Controls:

- **Moving through world:** By **pointing** the controller of choice in a specific direction and pressing the **grip button(s)** on the handle.
- **Adjust movement speed:** By swiping up- or downwards on a **trackpad** (Vive) or pressing up or down on a **joystick** (Quest 3) on one of the controllers.
- **Interaction with UI and other world elements:** By **pointing** at an element with one of the controllers and pressing the **trigger** on the back to interact with it.
- **Snapturn +/- 45 degrees:** By swiping left or right on a **trackpad** (Vive) or pressing left or right on a **joystick** (Quest 3) on one of the controllers.
- **Open menu:** By **rotating your hand** of choice sideways, **as if reading your watch** (advice to use recessive hand).
(Additionally, we could also implement the use of the **menu button** on the left-hand controller. A first press would open the menu, a second press would close it.)

UI Interaction:

1. Take a controller in each hand.
2. Open the menu by **rotating your hand** of choice sideways, **as if reading your watch** (advice to use recessive hand). The menu will open above said controller. The menu will remain open as long as you keep your hand in the correct position, even if the UI moves out of your field of view.
(Additionally, we could also implement the use of the **menu button** on the left-hand controller. A first press would open the menu, a second press would close it.)
3. Select one of the menu icons by **pointing** the second controller and pressing the **trigger** on the back (advice to use dominant hand).
4. **Data layers window:** Select one of the data layers from the list through point and click, as described above. This will open a second window on its right side which contains all the available display modes for said data layer.
5. **Specific data layer display modes window (example: wind farms):** Toggle on one or multiple display modes of interest through point and click. This will visualise the data in the world.

When applicable, for example in the case of a **heat map**, a display mode will be

accompanied by a legend.

When applicable, for example in the case of **area points** (wind turbine models), a display mode will be accompanied by a button that allows the user to cycle through the various phases. Each phase will visualise the model in a different state of construction. (In the future, this feature will be linked to the timeline.)

Each display mode can be separately collapsed or expanded through point and click.

Loss of Tracking:

If a controller loses tracking, an animated loading circle should appear (in the middle of the screen?) with a warning 'Reconnecting controller x'.

1.3 User Interface (UI)

The UI consists of two main elements: a left vertical bar and a horizontal bar at the top.

1.3.1.1 Left Vertical Bar:

- **PO Logo:** Clicking the PO logo prompts a menu with the following options:
 - **Resume:** Continue your current session.
 - **Credits:** View the credits for the application.
 - **Main Menu:** Return to the main menu.
 - **Exit Application:** Close the application.



Figure 3: Procedural Ocean Logo Icon

3D Models and Visualizations

- **Energy Constructions:** Displays 3D models of different types of energy constructions.
 - Show or hide cables on both seabed and surface visualizations.
 - Display or hide transformer stations and wind farm areas.



Figure 4: Energy Constructions Logo Icon

- **Fishing:**
 - Displays 3D models of four species of fish.
 - Show or hide the 3D models of the fish.
 - Display a heatmap of fish locations in the area. Note that some fish species may be more prevalent in certain areas.



Figure 4: Fishing Logo Icon

- **Shipping:**

- Contains two display options:
 - Shipping intensity heatmap.
 - Visibility of 3D models.



Figure 5: Shipping Logo Icon

- **Weather:**

- Controlled by a real-time API.
- Allows the user to change various weather parameters:
 - Time of day.
 - Cloud cover.
 - Wave height.
 - Wind speed.
 - Visibility.
 - Turbidity.
- Includes a reset button to revert weather conditions to their current real-time values.



Figure 6: Weather Logo Icon

- **Settings:**

- Master volume control.
- Graphics settings.
- Controls configuration.



Figure 7: Settings Logo Icon

- **Mini Map:**

- Displays the player's location in the North Sea.
- Can show various heatmaps indicating locations of different fish species.

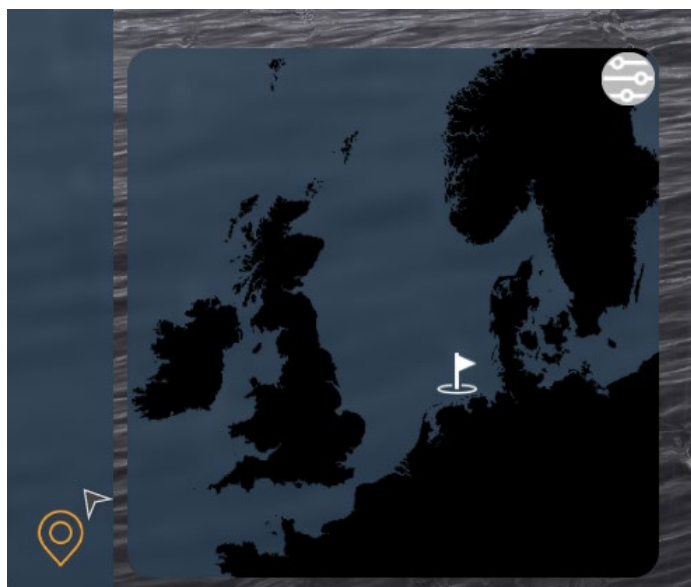


Figure 8: Mini Map

These elements provide a comprehensive and interactive environment, allowing users to customize their experience and obtain detailed information on various aspects of the simulation.

1.3.1.2 Horizontal Bar At The Top:

- **Time Control:** Manage the simulation speed with the following options:
 - **Slow Down:** Reduce the speed of the simulation.
 - **Normal:** Run the simulation at normal speed.
 - **Speed Up 2x:** Double the speed of the simulation.
 - **Speed Up 10x:** Increase the simulation speed by ten times.



Figure 9: Time Control UI

1.4 Player Interaction

While exploring the game, players can engage in various actions:

- **Select an Object:** Click on an object to display data about the selected object.
- **Dive:** Go underwater to explore what lies beneath the surface.
- **Build Wind Turbine:** Double-click on a location to construct a wind turbine there.
- **Maintenance Vessel:** Double middle mouse click to summon a maintenance vessel to perform maintenance on a wind turbine.
- **End of the Level:** A grid indicates the boundaries of the explorable area.

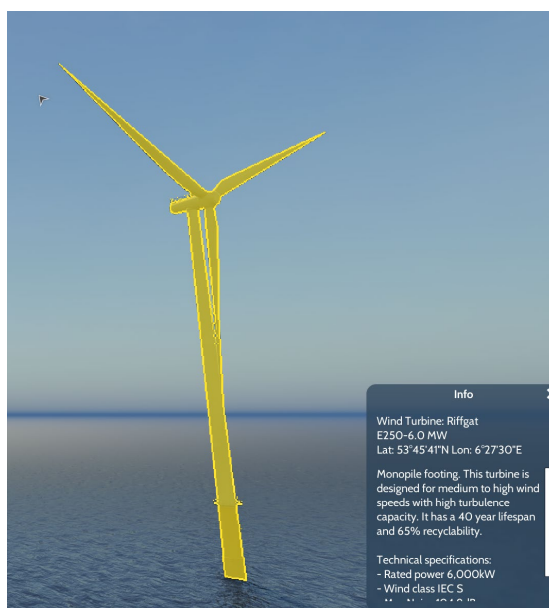


Figure 10: Player Interaction